DATASHEET

Promotion

AXS15231

In-Cell IC Integrates 540-channel 8-bit Source Driver and GIP Gate Driver and Touch Panel Controller Into a Single Chip with TP Controller Supports Real Multi-Touch Capability

AXS Confidential

Version V0.0
2022-04-26

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

Table of Contents

TABLE OF CONTENTS 2
REVISION HISTORY 3
GENERAL DESCRIPTION 4
1. FEATURES 5
1.1. DISPLAY 5
1.2. TOUCH 5
1.3. POWER 6
1.4. OTHERS 6
2. BLOCK DIAGRAM 7
2.1. BLOCK FUNCTION 7
2.1.1. Touch Function 7
2.1.2. Touch Operation Mode 8
2.1.3. MCU 8
2.1.4. Sleep In/Out Sequence 9
2.1.5. Deep Standby Mode enter/exit Sequence 10
2.1.6. System interface 10
2.1.7. Grayscale voltage generating circuit 10
2.1.8. Timing controller 10
2.1.9. Oscillator (OSC) 10
2.1.10. Source driver circuit 11
2.1.11. Gate driver circuit 11
2.1.12. LCD driving power supply circuit 11
3. PIN DESCRIPTION 12
3.1. POWER SUPPLY AND REGULATOR PINS 12
3.2. DRIVE INTERFACE LOGIC PIN 13
3.3. TP INTERFACE LOGIC PIN 16
3.4. MIPI INTERFACE 17
3.5. TEST/DUMMY SIGNAL/ OTHER 17
3.6. PANEL DRIVER SIGNALS 17
4. CHIP IMFORMATION 18
4.1. PAD ASSIGNMENT 18
4.2. PAD DIMENSION 19
4.3. PAD CENTER COORDINATE 20
5. DISCLAIMER 21

AXS Confidential

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

REVISION HISTORY

Date	Version	Description	Page	Author
2022/4/26	V0.0	Preliminary promotion version	35	paulson

AXS Confidential

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

GENERAL DESCRIPTION

The AXS15231 highly integrates a-Si TFT LCD driver and Super in-cell Touch controller, is a 16,777,216-color System-on-Chip (SOC) driver LSI designed for small and medium size TFT LCD display, and is capable of supporting up to 360RGBx640(Dual gate) pixels in resolution. The 540-channel source driver can provide true 8-bit resolution and generate 256 Gamma-corrected values with an internal D/A converter.

The AXS15231 is able to operate with low IO interface power supply. Incorporating with several charge pumps, the AXS15231 can generate various voltage levels by an on-chip power management system for gate and source driver. Moreover, PWM for LED backlight, wake up-button enabling, respiratory lights and other functions, to provide customers with better experience.

In addition, the external Flash of AXS15231 can store not only the firmware used for Touch controller, but also the Initial code of LCD driver. After loading the initial code through the external Flash, the HOST only needs to send out "Sleep out" and "Display on" to turn on the LCD.

The built-in timing controller in the AXS15231 can support several functions to meet a wide variety of requirements for portable display applications. It provides several system interfaces, including MIPI/QSPI/DSPI/SPI/RGB/MCU, which can be used to configure the system. Furthermore, it can also archive high speed display data transmission by using the MIPI interface.

AXS Confidential

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

1. Features

1.1. Display

1. One-chip solution for color amorphous TFT-LCD with incell
   Display Resolution
   - Dual gate: 360RGB(up to 360,2 lines/step) x 640(up to 640, 2 lines/step) in video mode
   - Single gate:180RGB(up to 180,2 lines/step) x 1024(up to 1024,2 lines/step) in video mode
   - Dual gate: 360RGB(up to 360,2 lines/step) x 420(up to 360, 2 lines/step) in command mode
   - Single gate:180RGB(up to 180,2 lines/step) x 850(up to 850,2 lines/step) in command mode
1. Support in-chip OTP
   - 2K*8 bit
   - OTP can stores trimming、ID1，ID2（factory ID，3bit）data
2. Analog gamma，256 gray level
3. VCOM: -2.5V~0V (10mv/step)
4. colour depth 888，666，565
5. Support 8-color and 2-color mode
6. 1 lane MIPI DSI:MIPI DSI2 V1.0 / V1.1 / V1.2, DCS：1.01.00 compatible，max speed 500Mbps
7. SPI/DSPI
8. QSPI（64Mbps/Lane）
9. Parallel 8080-series MCU Interface (8-bit, 9-bit, 16-bit, and 18-bit)
10. RGB
11. Column Inversion
12. Support 0 Flash

1.2. Touch

13. 8-bit embedded MCU
14. Support two-point detection
15. Point reporting rate 60Hz
16. Support single or double click gesture to wake up
17. Super self-capacitance detection technology
18. I2C/SPI data communication interface
19. Internal ESD detection
20. 50 SX channels
21. Power saving mode
22. VDDI_TP and VDDI supplied independently

AXS Confidential

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

1.3. Power

23. 2-power mode
    - VCI：3.0V~3.6V，typ ：3.3V
    - VDDI：1.65V ~ 3.6V，typ： 1.8V
24. 1-power mode
    - VCI/VDDI：3.0V~3.6V，typ ：3.3V

1.4. Others

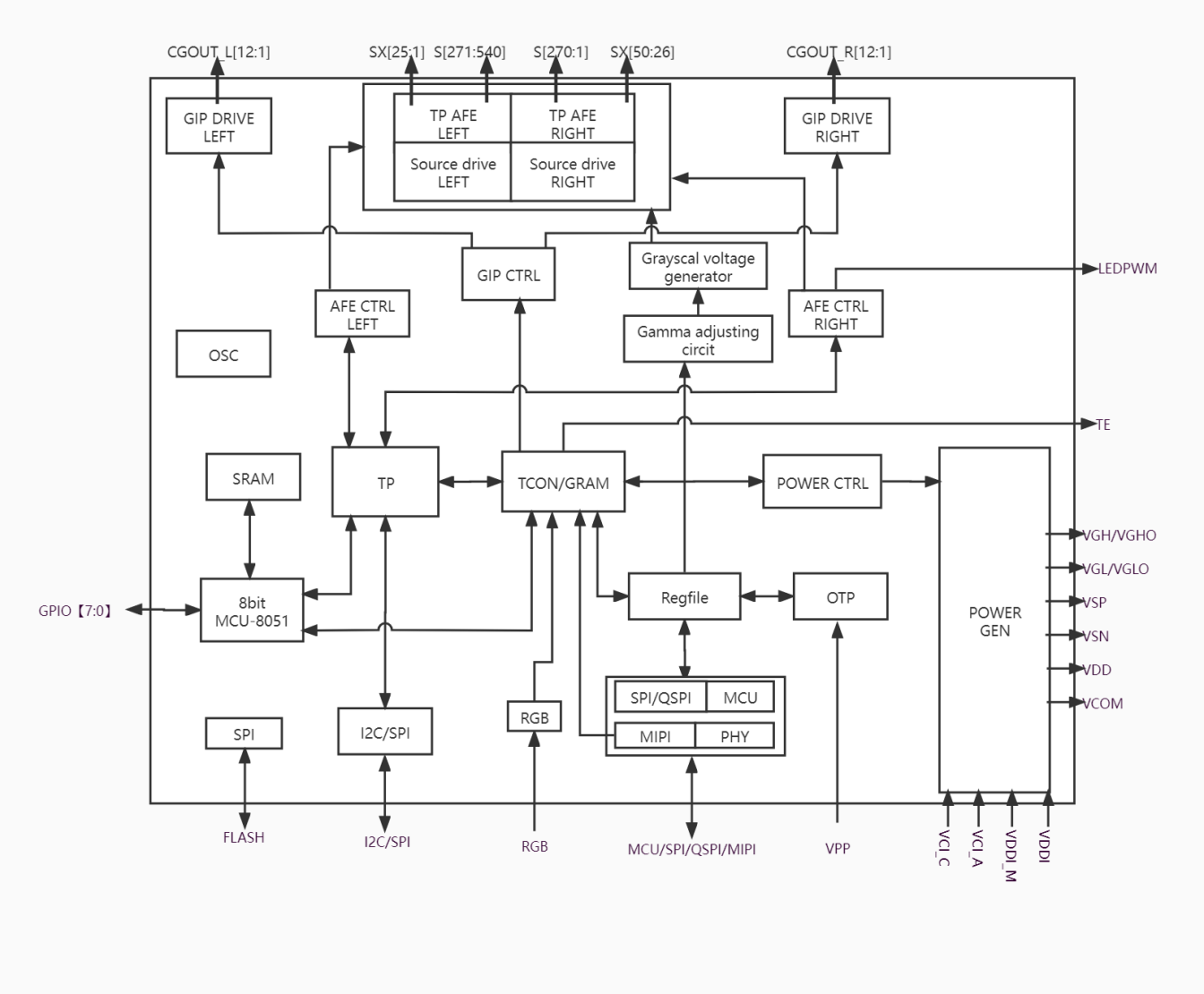
25. ESD
    - HBM≥2000V
    - MM≥200V
    - Latch up≥±200mA
26. COG Package

AXS Confidential

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

2. Block Diagram

2.1. Block Function



2.1.1. Touch Function

The touch part of AXS15231 mainly consists of the following components:

AFE Controller

AFE controller completes the scanning of the sensors in the touch panel, and sends the data of touch sensors after scanning to the MCU for data processing.

Embedded MCU

MCU and SOC subsystems complete the control, data processing, LCD operation and coordination,

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

HOST communication and other functions of the whole touch systems.

I2C/SPI serial interface

The Slave end of I2C/SPI in AXS15231 is the interface for touch communication with HOST. The control interface consists of two signals INT (is a GPIO, specified in the firmware) and RSTN (only one rstn-pin, shared by driver and touch). Whenever there is effective touch sensed on the touch screen, Touch controller will send data transfer request to the HOST via INT port, and complete the point report to the HOST. HOST can communicate with AXS15231 via I2C or SPI. HOST can also reset Touch controller through RSTN port.

External Flash

External Flash, used to store the Firmware, and LCD initialization code, can be added into the Touch controller.

Watchdog

Watchdog is used to ensure the stability of the chip when in operation

Internal voltage regulator

Internal voltage regulator generates 1.6V power supply, which is to provide power to logic circuit.

2.1.2. Touch Operation Mode

Touch controller has the following three operation modes:

Normal operating mode

In this mode, Touch controller scans the screen, and detects the touch actions.

Monitor Mode

In this mode, Touch controller scans the screen intermittently to save power.

Sleep mode

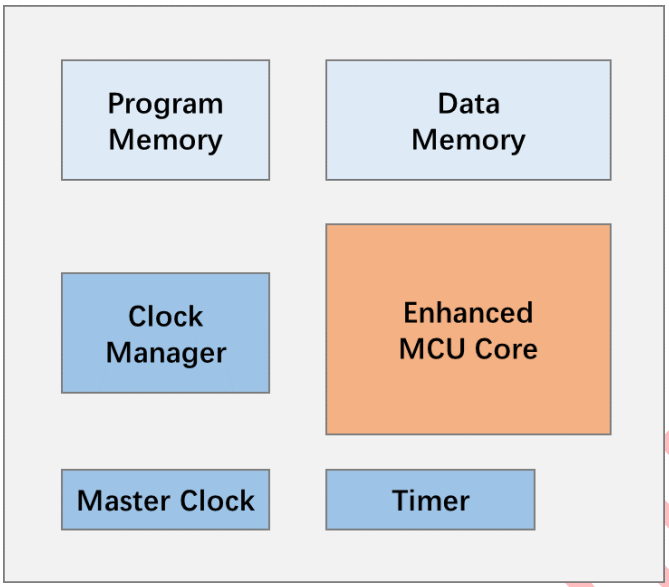
In this mode, Touch controller enters ultra low-power standby mode, HOST can only wake-up Touch controller via external signal to enter the normal operating mode. The power consumption in this mode is extremely small, and can greatly extend the standby time of mobile portable devices.

2.1.3. MCU

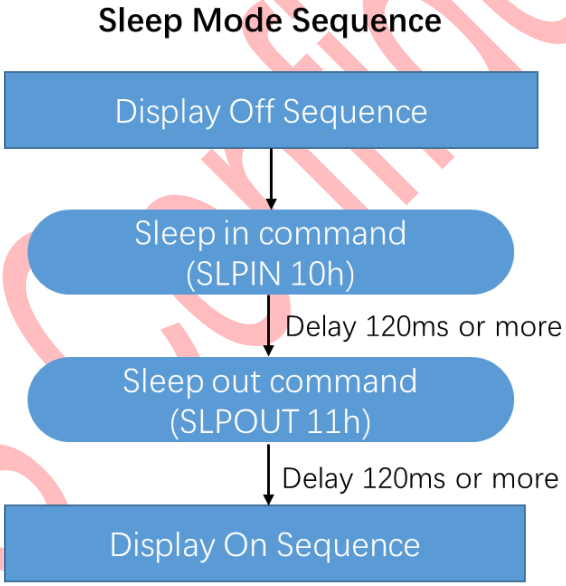
This section describes some critical features and operations supported by the 8051 compatible MCU.The figure below shows the overall structure of the MCU block. In addition to the 8051 compatible MCU core, we have added the following circuits.

AXS Confidential

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.



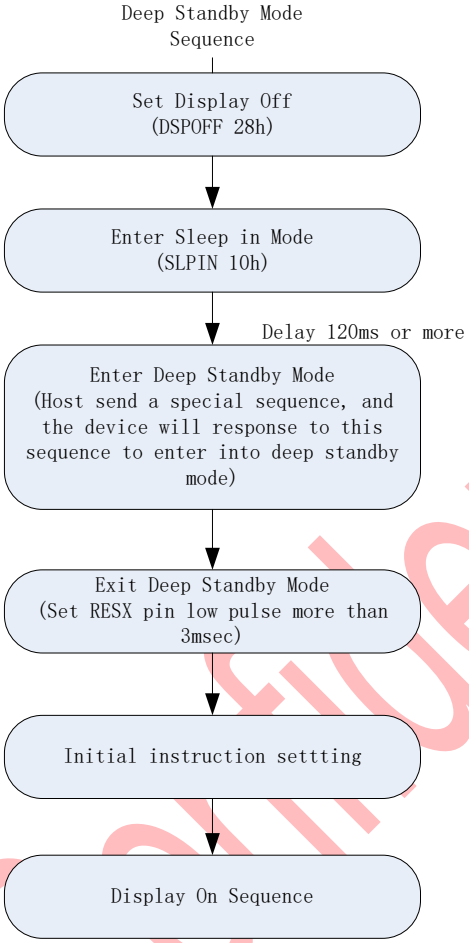
2.1.4. Sleep In/Out Sequence



AXS Confidential

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

2.1.5. Deep Standby Mode enter/exit Sequence



2.1.6. System interface

The AXS15231 supports the video data transmitted through the high-speed system interface, MIPI (Mobile Industry Processor Interface).

2.1.7. Grayscale voltage generating circuit

AXS15231 has true 8-bit resolution D/A converter, digital gamma cooperates with OP-AMP structure to enhance display quality. The grayscale voltage can be adjusted by grayscale data set in the γ-correction register and RGB can be adjusted separately.

2.1.8. Timing controller

AXS15231 has a timing controller, which can generate a timing signal for internal circuit operation such as gate output timing, image data accessing timing, etc.

2.1.9. Oscillator (OSC)

The AXS15231 has an internal oscillator to generate system clock.

AXS Confidential

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

2.1.10. Source driver circuit

AXS15231 consists of a 540-output source driver circuit.

2.1.11. Gate driver circuit

AXS15231 consists of a gate driver control circuit. The gate driver circuit outputs gate driver signals level at either VGH or VGL level.

2.1.12. LCD driving power supply circuit

The LCD driving power supply circuit generates the voltage levels VDD, VSP, VSN, VGH, VGL, VCOM for driving a LCD. All these voltages can be adjusted by register setting.

AXS Confidential

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

3. Pin Description

3.1. Power Supply and Regulator pins

Name	I/O	Default value	Description
VCI_A	I	3.0V ~ 3.6V，typ:3.3V	External input voltage
VCI_C	I	3.0V ~ 3.6V，typ:3.3V	External input voltage
VDDI	I	1.65V ~ 3.6V，typ:1.8V	External input voltage
VDDI_DRV	I	1.65V ~ 3.6V，typ:1.8V	External input voltage
VDDI_M	I	1.65V ~ 3.6V，typ:1.8V	External input voltage for mipi-ldo
VFLASH	I	1.65V ~ 3.6V，typ:1.8V	External input voltage for external spi-flash
VSP	O	5.0V ~ 7.0V，typ:6.0V	Charge pump output voltage or External input voltage
VSN	O	-5.4V ~ -4.5V，typ:-5.0V	Charge pump output voltage or External input voltage
VCOM	O	-2.5V ~ 0V，typ: 0V	Regulator output voltage
VDD	O	1.4V ~ 1.7V，typ:1.6V	Internal LDO output for digital power supply
VSSD	I	GND	Digital ground
VSSA_1	I	GND	Analog ground, connect to VSSD on FPC
VSSA_2	I	GND	Analog ground, connect to VSSD on FPC
VSSA_C	I	GND	Analog ground, connect to VSSD on FPC
VSSA_M	I	GND	MIPI ground, connect to VSSD on FPC
VPP	I	8.0V ~ 8.5V	supply for OTP program

AXS Confidential

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

Name	I/O	Default value	Description
VDN	O	1.4V ~ 1.7V, typ:1.6V	Internal LDO output
VGH_R	O	11V ~ 15.5V	14V power supply for GIP positive, connect to VGH_L on FPC
VGH_L	O	11V ~ 15.5V	14V power supply for GIP positive, connect to VGH_R on FPC
VGHO1	O	10V ~ 14.5V	External an 100nf capacitor
VGL	O	-10.5V ~ -7.5V	-10V power supply for GIP negtive. VGL is also the substrate voltage of the whole chip
VGLO1	O	-10V ~ -7.5V	External an 100nf capacitor
VDDA_TP	O	VCI or VSP	TP power, connect to VCI/VSP on FPC
VREF_TP	O	Power Output	Regulator output voltage for TP-AFE
VCG_TP	O	2.0V ~ 5.0V	Regulator output voltage for TP-AFE
VCOM_PASS_R	O	0 ~ 5.0V	Pulse signals synchronize with tp scan/hiz/gnd
VCOM_PASS_L	O	0 ~ 5.0V	

3.2. Drive interface logic pin

Name	I/O	Default value	Description
RSTN	I	1,pull up without control	This signal will reset the device and must be applied to properly initialize the chip.Signal is active low.pull up to avoid floating. Caution: TP and Driver share the same RSTN-PIN
LED_PWM	O	0,50pf	Backlight control setting pin.Output load 50pf
TE	O	0,50pf	output tearing effect signal from IC to host
IM[3:0]	I	0,pull down without control	select the input interface 1010:QSPI 1001:spi 3 wire 9bits dual data lane from driver spi 1000:dbi 18bits; 0111:dbi 16bits; 0110:dbi 9bits; 0101:dbi 8bits; 0100:spi 4Wire 8bits dual data lane; 0011:spi 3wire 9bits+rgb

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.
AXS Confidential

Name	I/O	Default value	Description
			0010:spi 4wire 8bits+rgb 0001:spi 3wire 9bits dual data lane from gpio 0000:mipi
CSX	I	1,pull up without control	Chip select input pin("Low" enable) in SPI slave ,pull up to avoid floating. This pin is not used for MIPI I/F, please connect to VDDI.
SCL	I	1,pull up without control	Synchronous clock signal in SPI slave, pull up to avoid floating. This pin is not used for MIPI I/F, please connect to VDD.
RS	I	0,pull down without control	command or parameter selection in spi 4wire 8bits.
DIN_SDA	B	0,pull down without control	The bidirectional data pin of spi slave. Default input. pull down to avoid floating. Output load 50pf.This pin is not used for MIPI I/F, please connect to VSSD. In QSPI mode, it used as QSPI_DIN0 In SPI 3wire/4wire mode-0, it used as SPI_SDA In SPI 3wire/4wire mode-1, it used as MISO
DIN_SDA_DUAL	I	0,pull down without control	The second data input pin in spi dual data lane of spi slave.Default input.pull down to avoid floating.Output load 50pf.This pin is not used for MIPI I/F,please connect to VSSD. In QSPI mode, it used as QSPI_DIN1 In SPI 3wire/4wire mode-1, it used as MOSI
VSYNC_QSPI_DIN2	I	1,pull up without control	Frame synchronizing signal of rgb interface. Used as csn of 8080 mcu interface for command mode. In QSPI mode, it used as QSPI_DIN2
HSYNC	I	1,pull up without control	Line synchronizing signal of rgb interface. Used as d/cx of 8080 mcu interface for command mode.
DE_QSPI_DIN3	I	0,pull down without control	Data enable signal of rgb interface. Used as wr of 8080 mcu interface for command mode. In QSPI mode, it used as QSPI_DIN3.
PCLK	I	0,pull down without control	pixel clock signal of rgb interface. Used as rd of 8080 mcu interface for command mode.
DB[17:0]	I	0,pull down without control	DB[17:0] input data of rgb interface. DB[17:0] of 8080 mcu interface for command mode.

IM3	IM2	IM1	IM0	Interface	Connect Pin
0	0	0	0	MIPI	HS_DN,HS_DP,HS_CN,HS_CP,RSTN
0	0	0	1	spi 3wire 9bits dual data lane from gpio	DIN_SDA,DIN_SDA_DUAL,SCL,CSX,RSTN
0	0	1	0	SPI 4wire/8bits RGB	DIN_SDA(MISO),DIN_SDA_DUAL(MOSI),RS(D/CX),SCL,CSX,RSTN
0	0	1	1	spi 3wire 9bits+rgb 16/18/24-bit	Video Stream Interface:DB[15/17/23:0],VSYNC_QSPI_DIN2,HSYNC,DE_QSPI_DIN3,PCLK Control Interface:DIN_SDA(MISO),DIN_SDA_DUAL(MOSI),SCL,CSX,RSTN
0	1	0	0	SPI 4wire	DIN_SDA,DIN_SDA_DUAL,RS(D/CX),SCL,CSX,RSTN

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

				8bits dual data lane	
0	1	0	1	MCU 8080 8bits	DB[7:0],VSYNC_QSPI_DIN2(CSX),HSYNC(D/CX),DE_QSPI_DIN3(WRX),PCLK(RDX),RSTN
0	1	1	0	MCU 8080 9bits	DB[8:0],VSYNC_QSPI_DIN2(CSX),HSYNC(D/CX),DE_QSPI_DIN3(WRX),PCLK(RDX),RSTN
0	1	1	1	MCU 8080 16bits	DB[15:0],VSYNC_QSPI_DIN2(CSX),HSYNC(D/CX),DE_QSPI_DIN4(WRX),PCLK(RDX),RSTN
1	0	0	0	MCU 8080 18bits	DB[17:0],VSYNC_QSPI_DIN2(CSX),HSYNC(D/CX),DE_QSPI_DIN5(WRX),PCLK(RDX),RSTN
1	0	0	1	SPI 3wire 9 bits dual lane from drv SPI	DIN_SDA,DIN_SDA_DUAL,SCL,CSX,RSTN
1	0	1	0	QSPI	DE_QSPI_DIN3,VSYNC_QSPI_DIN2,DIN_SDA,DIN_SDA_DUAL,SCL,CSX,RSTN

AXS Confidential

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

3.3. TP interface logic pin

Name	I/O	Default value	Description
RSTN	I	weak pull up without control	This signal will reset the device and must be applied to properly initialize the chip.Signal is active low.pull up to avoid floating. Caution: TP and Drive share the same RSTN-PIN.
WORK_MODE	I	weak pull down without control	Selection of work mode,high represent 0 flash mode;low represent extern flash. need weak pull down to avoid floating.
GPIO[7:0]	B	open drain,IC internal pull up,4.7K;	work for I2C interface clock or data as I2C master,when WORK_MODE=0;GPIO[1:0] only used for I2C slave interface; GPIO[7:4] can used for spi master interface of flash or other function.
		50pf	work for spi interface or test output, when WORK_MODE=1,GPIO[7:4] only used for spi slave interface.
		when output,50pf;when input ,weak pull down and pull up with control	when output,50pf;when input ,weak pull down and pull up with control
TS_SEL	I	Test signal	test pad
OSC_TE	I	weak pull down without control	osc select;1:extern osc input.0:internal osc clock
OSC_IN	I	weak pull down without control	extern input osc clock
SCAN_MODE	I	weak pull down without control	selection of scan chain mode,active high.
SCAN_EN	I	weak pull down without control	enable of scan chain
SCAN_IN[2:0]	I	weak pull down without control	input data of scan chain
SCAN_OUT[2:0]	O	50pf	output data of scan chain
SX[50:1]	O	-	TP Sensor Pins

WORK_MODE	Interface	Connect Pin
0	I2C	GPIO[0] : SCL GPIO[1] : SDA GPIO[4] : FLASH_CS GPIO[5] : FLASH_SCK GPIO[6] : FLASH_MOSI GPIO[7] : FLASH_MISO
1	SPI	GPIO[4] : CS_SLAVE GPIO[5] : SCK_SLAVE GPIO[6] : MOSI_SLAVE GPIO[7] : MISO_SLAVE

AXS Confidential

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

3.4. MIPI Interface

Name	I/O	Default value	Description
HS_CP	I	MIPI Input	MIPI-DSI clock Lane positive-end input pin
HS_CN	I	MIPI Input	MIPI-DSI clock Lane negative-end input pin
HS_DP	I/O	MIPI Input /Output	MIPI-DSI data Lane 0 positive-end input/output pin
HS_DN	I/O	MIPI Input /Output	MIPI-DSI data Lane 0 negative-end input/output pin
SMT_MIPI	I	0,pull down without control	Selection the mipi low power mode.

3.5. Test/Dummy Signal/ Other

Name	I/O	Default value	Description
DUMMY	O	-	DUMMY PIN, no practical application
ATEST[5:0]	I/O	Test signal	test pad
BGR_CORE_SEL_PAD	I	Test signal	test pad
POR_MODE_SEL_PAD	I	Test signal	test pad
VREFN	O	Test signal	test pad
VBG_PAD	O	Test signal	test pad
POCN_VCI_PAD	O	Test signal	test pad
RF_CHECK	I	-	Detect RF interference

3.6. Panel driver Signals

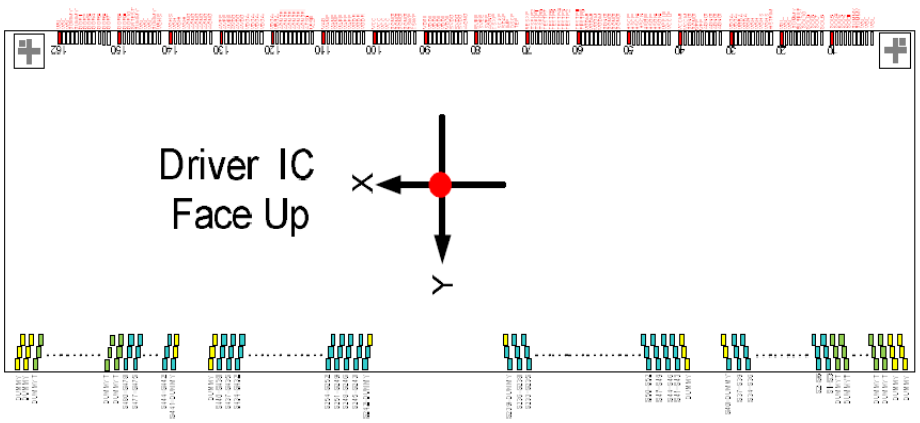
Name	I/O	Default value	Description
CGOUT_L[12:1]	O	GIP Output	Gate control signals for panel in left side of IC
CGOUT_R[12:1]	O	GIP Output	Gate control signals for panel in right side of IC
S[540:1]	O	Source driver Output	Output source driver signals. The D/A converted 256-gray-scale analog voltage output.
			Source output mapping with different resolution.
			360 RGB x (360, others), (Source output from S1 to S540)

AXS Confidential

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

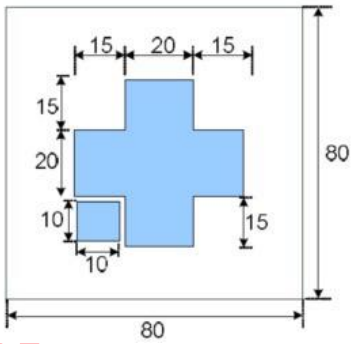
4. CHIP IMFORMATION

4.1. PAD Assignment

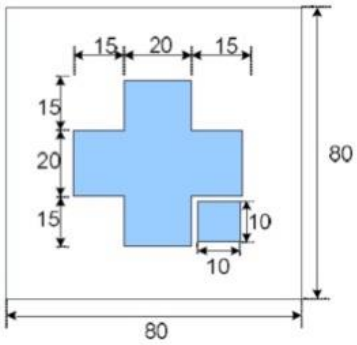


Chip size	13540um×930um
Input bump size	30um×75um
Output bump size	15um×75um
Mark left(x,y)	(-7190,352)
Mark right(x,y)	(6190,352)

●Alignment Mark Left:



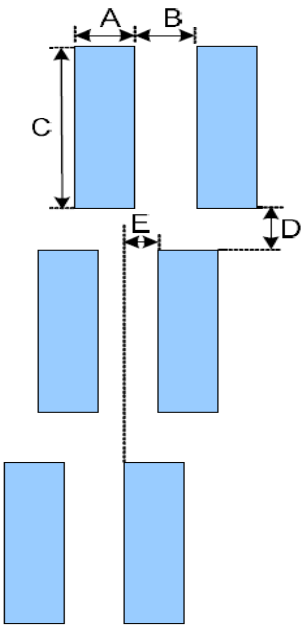
●Alignment Mark Right:



The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

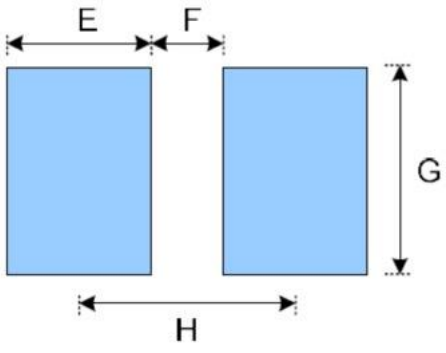
4.2. PAD Dimension

●Output Pads



Symbol	Item	Size
A	Bump Width	15 um
B	Bump Gap 1 (Horizontal)	18 um
C	Bump Height	75 um
D	Bump Gap 2 (Vertical)	25 um
E	Bump Pitch	11um

●Input Pads



Symbol	Item	Size
E	Bump Width	30 um
F	Bump Gap	15 um
G	Bump Height	75 um
H	Bump Pitch	45 um

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

4.3. PAD CENTER COORDINATE

AXS Confidential

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

5. DISCLAIMER

The information appearing in this publication is believed to be accurate.Integrated circuits sold by AIXIESHENG covered by the warranty and patent indemnification provisions stipulated in the terms of sale only. AIXIESHENG Technology makes no warranty, express, statutory implied or by description regarding the information in this publication or regarding the freedom of the described chip(s) from patent infringement. Furthermore, AIXIESHENG Technology makes no warranty of merchant ability or fitness for any purpose. AIXIESHENG Technology reserves the right to halt production or alter the specifications and prices at any time without notice. Accordingly, the reader is cautioned to verify that the data sheets and other information in this publication are current before placing orders. Products described herein are intended for use in normal commercial applications. Applications involving unusual environmental or reliability requirements, e.g. military equipment or medical life support equipment, are specifically not recommended without additional processing by AIXIESHENG Technology for such applications. Please note that application circuits illustrated in this document are for reference purposes only.

AXS Confidential

The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.